Khulna University of Engineering & Technology

MATH 2123 - Mathematics III

2018 Regular Examination

Beginner-Friendly Topic Guide + Full Step-by-Step Solutions

Question set	Main topics
1	Matrix definitions, symmetric/skew decomposition, inverse by row operations
2	Rank, echelon/canonical/normal forms, P and Q, skew-symmetric property
3	Consistency of systems, parameter cases, non-trivial solutions, trace identity
4	Elementary/equivalent matrices, involutory/idempotent, eigenvalues/eigenvectors
5	Laplace existence, special transforms, convolution theorem
6	Laplace solution of ODE, fixed-fixed beam deflection
7	Scalar triple product, moment of force, solenoidal/conservative fields
8	Orthogonal surfaces, Green's theorem, surface flux on a cylinder

How to use this guide: Read the "Basics" first, learn the repeatable procedure, then study the worked solution. The purpose is not to memorize one answer, but to learn a method that can be reused in similar exam questions.

Exam tip: For a 3-hour exam, first identify the question type. Matrix and Laplace questions are usually method-driven: if you write the correct standard steps clearly, you can collect substantial marks even before the final answer.

Question Set 1 - Core Matrix Concepts and Inverse

Topics

• Orthogonal, Hermitian, idempotent and lower triangular matrices

• Decomposing a matrix into symmetric and skew-symmetric parts

• Finding an inverse by elementary row transformations

Basics you need

For a real square matrix, transpose means interchanging rows and columns. For a complex matrix, the conjugate transpose is written A* (or A^H). A matrix is invertible only when det(A) is not zero.

```
Symmetric:       S^T = S
Skew-symmetric:  K^T = -K
Any square A:    A = (A+A^T)/2 + (A-A^T)/2
Inverse method:  [ A | I ]  --row operations-->  [ I | A^(-1) ]
```

1(a) Definitions with examples

Orthogonal matrix

A real square matrix Q is orthogonal if Q^T Q = QQ^T = I. Therefore Q^(-1) = Q^T.

```
Example: Q = [ 0  -1 ]
             [ 1   0 ]
Then Q^T Q = I.
```

Hermitian matrix

A complex square matrix H is Hermitian if H* = H, where * means conjugate transpose.

```
Example: H = [ 2   i ]
             [ -i  3 ]
```

Idempotent matrix

A square matrix P is idempotent if P^2 = P.

```
Example: P = [ 1  0 ]
             [ 0  0 ]
```

Lower triangular matrix

A square matrix is lower triangular when every entry above the main diagonal is zero.

```
Example: L = [ 1  0  0 ]
             [ 2  3  0 ]
             [ 4  5  6 ]
```

1(b) Express A as symmetric + skew-symmetric

```
A = [ 1   1   3 ]
    [ 5   2   6 ]
    [ 2  -1  -3 ]
```

Problem-solving procedure

Step 1. Compute A^T.

Step 2. Use S=(A+A^T)/2.

Step 3. Use K=(A-A^T)/2.

Step 4. Check S^T=S, K^T=-K, and S+K=A.

```
A^T = [ 1   5   2 ]
      [ 1   2  -1 ]
      [ 3   6  -3 ]

S = (A+A^T)/2
```

```
  = [ 1    3    5/2 ]
    [ 3    2    5/2 ]
    [ 5/2  5/2  -3  ]

K = (A-A^T)/2
  = [ 0    -2    1/2 ]
    [ 2     0    7/2 ]
    [ -1/2 -7/2   0  ]
```

Final answer: A = S + K with the symmetric matrix S and skew-symmetric matrix K shown above.

1(c) Inverse by elementary row transformations

```
A = [ -2   1   3 ]
    [  0  -1   1 ]
    [  1   2   0 ]
```

Problem-solving procedure

Step 1. Write the augmented matrix [A | I].

Step 2. Create pivots 1,0,0 down the left side.

Step 3. Eliminate above and below each pivot until the left block is I.

Step 4. Read A^(-1) from the right block.

```
Start:
[ -2  1  3 | 1 0 0 ]
[  0 -1  1 | 0 1 0 ]
[  1  2  0 | 0 0 1 ]

R1 <-> R3
R3 <- R3 + 2R1
R2 <- -R2
R3 <- R3 - 5R2
R3 <- (1/8)R3
R2 <- R2 + R3
R1 <- R1 - 2R2

Result:
[ 1 0 0 | -1/4   3/4   1/2 ]
[ 0 1 0 |  1/8  -3/8   1/4 ]
[ 0 0 1 |  1/8   5/8   1/4 ]
```

A^(-1) = [[-1/4, 3/4, 1/2], [1/8, -3/8, 1/4], [1/8, 5/8, 1/4]].

Exam tip: When using Gauss-Jordan, write every row operation beside the matrix. It makes the method easy to follow and protects marks even if a later arithmetic slip occurs.

Question Set 2 - Rank, Canonical Form and Normal Form

Basics you need

Rank is the number of linearly independent rows (or columns). In row-echelon form it equals the number of nonzero rows/pivots. A row canonical form is the reduced row-echelon form (RREF). A normal form under both row and column operations is [I_r 0; 0 0], where r is the rank.

2(a) Echelon form, canonical form, normal form and rank

```
A = [  2  1  -1   3 ]
    [ -3  0   1  -2 ]
    [ -4  5   0  -1 ]
```

Problem-solving procedure

Step 1. Use row operations to produce zeros below each pivot.

Step 2. Count pivots to get the rank.

Step 3. Continue to RREF for canonical form.

Step 4. Use column operations to remove non-pivot columns for normal form.

```
R2 <- 2R2 + 3R1
R3 <- R3 + 2R1
R3 <- 3R3 - 7R2

Echelon form:
[ 2  1  -1    3 ]
[ 0  3  -1    5 ]
[ 0  0   1  -20 ]

Now:
R2 <- R2 + R3, then R2 <- (1/3)R2
R1 <- R1 + R3
R1 <- R1 - R2, then R1 <- (1/2)R1

Canonical (RREF):
[ 1  0  0   -6 ]
[ 0  1  0   -5 ]
[ 0  0  1  -20 ]

Column operation:
C4 <- C4 + 6C1 + 5C2 + 20C3

Normal form:
[ 1 0 0 0 ]
[ 0 1 0 0 ]
[ 0 0 1 0 ]
```

Rank(A) = 3.

2(b) Find nonsingular P and Q so that PAQ is normal form

```
A = [  2  -1   3   4 ]
    [ -1  -3   0   1 ]
    [  1  -2  -1   2 ]
```

A convenient shortcut is to use the first three columns, which form a nonsingular 3x3 matrix B. Let P=B^(-1). Then PA has I in its first three columns.

```
B = [  2  -1   3 ]
    [ -1  -3   0 ]
    [  1  -2  -1 ]
```

```
det(B)=22 != 0
```

```
P = B^(-1) = (1/22)[ 3  -7   9 ]
                   [ -1 -5  -3 ]
                   [ 5   3  -7 ]

PA = [ 1 0 0   23/22 ]
     [ 0 1 0  -15/22 ]
     [ 0 0 1    9/22 ]
```

Choose Q to subtract the last column combination from itself:

```
Q = [ 1 0 0  -23/22 ]
    [ 0 1 0   15/22 ]
    [ 0 0 1   -9/22 ]
    [ 0 0 0      1   ]

det(Q)=1 != 0

Therefore:
PAQ = [ 1 0 0 0 ]
      [ 0 1 0 0 ]
      [ 0 0 1 0 ]
```

One valid pair is the P and Q above. Many other valid nonsingular P,Q pairs are possible.

2(c) Why diagonal entries of a skew-symmetric matrix are zero

For a skew-symmetric matrix A, A^T=-A. Looking at the i-th diagonal entry gives a_ii = -a_ii. Therefore 2a_ii=0, so a_ii=0 for every i.

Every diagonal element of a real skew-symmetric matrix is zero.

Question Set 3 - Consistency of Linear Systems and Trace

Basics you need

```
For AX=b:
rank(A) = rank([A|b]) = number of unknowns  -> unique solution
rank(A) = rank([A|b]) < number of unknowns  -> infinitely many solutions
rank(A) < rank([A|b])                       -> no solution
```

3(a) Parameter a: no solution / unique / infinite solutions

```
x + y - z = 1
2x + 3y + a z = 3
x + a y + 3z = 2
```

Coefficient matrix:

```
A = [ 1  1  -1 ]
    [ 2  3   a ]
    [ 1  a   3 ]

det(A) = -(a-2)(a+3)
```

Problem-solving procedure

Step 1. If det(A) != 0, there is a unique solution.

Step 2. Check the exceptional values a=2 and a=-3 by comparing ranks.

Step 3. Classify each case.

```
Case 1: a != 2,-3
 det(A) != 0 -> unique solution.
 In fact: x=1, y=1/(a+3), z=1/(a+3).

Case 2: a=2
 rank(A)=rank([A|b])=2 < 3
 -> infinitely many solutions.

Case 3: a=-3
 rank(A)=2 but rank([A|b])=3
 -> no solution.
```

No solution: a=-3. Infinite solutions: a=2. Unique (nonzero) solution: a != 2,-3.

3(b) Check consistency and find non-trivial solutions

```
2x1 - 3x2 + 3x3 + 5x4 =  2
-x1      - 2x3 +  x4      =  3
3x1 + 2x2 -  x3 + 2x4      = -1
```

Row reduction of the augmented matrix gives:

```
[ 1 0 0   35/23 |  11/23 ]
[ 0 1 0  -44/23 | -48/23 ]
[ 0 0 1  -29/23 | -40/23 ]
```

There are 3 pivots for 4 unknowns, and rank(A)=rank([A|b])=3, so the system is consistent with one free variable. Let x4=t.

```
x1 = 11/23 - (35/23)t
x2 = -48/23 + (44/23)t
x3 = -40/23 + (29/23)t
x4 = t,        t is any real number.
```

The system is consistent and has infinitely many (non-trivial) solutions given by the parameter t above.

3(c) Show trace(AA^T)=trace(A^T A)

Let A be the displayed 2x3 matrix with entries a_ij. The diagonal entries of AA^T are row-wise sums of squares; the diagonal entries of A^T A are column-wise sums of squares.

```
trace(AA^T)
= (a11^2+a12^2+a13^2) + (a21^2+a22^2+a23^2)

trace(A^T A)
= (a11^2+a21^2) + (a12^2+a22^2) + (a13^2+a23^2)

Both expressions are the same total sum of squares.
```

Therefore trace(AA^T) = trace(A^T A).

Exam tip: For consistency questions, determinant is useful only for square systems. Rank comparison is the universal test.

Question Set 4 - Elementary Matrices, Involutory Matrices and Eigenvalues

4(a) Elementary and equivalent matrices

Elementary matrix: a matrix obtained by performing exactly one elementary row operation on an identity matrix. Multiplying E A performs the same row operation on A.

```
Example: swap R1 and R2 in I2:
E = [ 0 1 ]
    [ 1 0 ]
```

Equivalent matrices: A and B are equivalent if one can be obtained from the other through a finite sequence of elementary row and/or column operations. Equivalently, B=PAQ for nonsingular P and Q.

```
Example:
A = [1 2; 3 4]
R2 <- R2 - 3R1 gives B = [1 2; 0 -2].
So A and B are row-equivalent (hence equivalent).
```

4(b) If A is involutory, show 1/2(I+A) and 1/2(I-A) are idempotent

Involutory means A^2=I. Define P=(I+A)/2 and Q=(I-A)/2.

```
P^2 = (1/4)(I+A)^2
    = (1/4)(I + 2A + A^2)
    = (1/4)(2I + 2A)
    = (1/2)(I+A) = P.

Q^2 = (1/4)(I-A)^2
    = (1/4)(I - 2A + A^2)
    = (1/4)(2I - 2A)
    = (1/2)(I-A) = Q.
```

Both (I+A)/2 and (I-A)/2 are idempotent.

4(c) Eigenvalues and eigenvectors

Eigenvalue/eigenvector: for a square matrix B, a nonzero vector v is an eigenvector with eigenvalue lambda if Bv=lambda v, or equivalently (B-lambda I)v=0.

Three useful properties: (1) sum of eigenvalues = trace (with multiplicity), (2) product of eigenvalues = determinant, (3) if lambda is an eigenvalue of B then lambda^n is an eigenvalue of B^n.

```
B = [ 1  -3   3 ]
    [ 3  -5   3 ]
    [ 6  -6   4 ]

Characteristic equation:
(lambda - 4)(lambda + 2)^2 = 0

Eigenvalues: lambda=4 and lambda=-2 (double).
```

For lambda=4, solving (B-4I)v=0 gives a convenient eigenvector v=(1,1,2)^T.

For lambda=-2, the eigenspace has two independent vectors, for example (1,1,0)^T and (-1,0,1)^T.

Eigenvalues: 4, -2, -2. One eigenvector for lambda=4 is (1,1,2)^T.

Question Set 5 - Laplace Transform and Convolution

Basics you need

```
L{f(t)} = integral_0^infinity e^(-st) f(t) dt
L{integral_0^t f(u)du} = F(s)/s
L{t f(t)} = -dF/ds
Convolution: L^(-1){F(s)G(s)} = integral_0^t f(u)g(t-u)du
```

5(a) Existence conditions and L{1/sqrt(t)}

A common sufficient theorem says: if f is piecewise continuous on every finite interval and is of exponential order as t approaches infinity, then its Laplace transform exists for sufficiently large s. These are sufficient, not necessary conditions.

Here f(t)=t^(-1/2) is unbounded at t=0, so the ordinary sufficient theorem does not directly apply there. But the improper Laplace integral converges:

```
L{t^(-1/2)}
= integral_0^infinity e^(-st) t^(-1/2) dt
= Gamma(1/2) / s^(1/2)
= sqrt(pi) / sqrt(s),    s>0.
```

L{1/sqrt(t)} = sqrt(pi/s), for s>0. The transform exists.

5(b) Laplace transform of g(t)

```
g(t) = integral_0^t (sin u)/u du + t e^(2t) cos(2t)
```

First term: L{sin t / t}=tan^(-1)(1/s), so integrating from 0 to t divides the transform by s.

Second term: let F(s)=L{e^(2t)cos2t}=(s-2)/[(s-2)^2+4]. Then L{t f(t)}=-F'(s).

```
L{g(t)} = (1/s) tan^(-1)(1/s)
          + [ (s-2)^2 - 4 ] / [ ((s-2)^2 + 4)^2 ]

Overall region: s>2.
```

G(s) = tan^(-1)(1/s)/s + ((s-2)^2-4)/(((s-2)^2+4)^2).

5(c) Inverse Laplace by convolution

```
Find L^(-1){ 1 / [s^2 (s+1)^2] }.

1/s^2        <->  t
1/(s+1)^2    <->  t e^(-t)
```

By convolution:

```
f(t) = integral_0^t (t-u) [u e^(-u)] du
     = t - 2 + (t+2)e^(-t).
```

L^(-1){1/[s^2(s+1)^2]} = t - 2 + (t+2)e^(-t).

Exam tip: In convolution questions, first split the transform into two factors whose inverse transforms you already know. That is usually the hardest strategic step.

Question Set 6 - Laplace ODE and Beam Deflection

6(a) Solve the ODE by Laplace transform

```
x'' - 2x' + x = e^t,
x(0)=2,  x'(0)=-1.
```

Problem-solving procedure

Step 1. Take Laplace transforms term by term.

Step 2. Insert initial conditions immediately.

Step 3. Solve algebraically for X(s).

Step 4. Rewrite X(s) in standard inverse-transform forms.

```
L{x''} = s^2 X - 2s + 1
L{x'}  = sX - 2

(s^2 X - 2s + 1) - 2(sX-2) + X = 1/(s-1)

(s-1)^2 X - 2s + 5 = 1/(s-1)

X = 2/(s-1) - 3/(s-1)^2 + 1/(s-1)^3.
```

```
Using
L^(-1){1/(s-a)}   = e^(at)
L^(-1){1/(s-a)^2} = t e^(at)
L^(-1){1/(s-a)^3} = (t^2/2)e^(at),

x(t) = e^t [ 2 - 3t + t^2/2 ].
```

x(t) = e^t (t^2/2 - 3t + 2).

6(b) Fixed-fixed beam with triangular load on the left half

The beam has length L and is embedded (fixed) at both ends. The load is

```
w(x) = w0(1 - 2x/L),   0 <= x <= L/2
       0,                L/2 <= x <= L
```

For an Euler-Bernoulli beam with constant EI, use EI y^(4)(x)=w(x), taking positive y in the load direction. Fixed ends give y(0)=y'(0)=y(L)=y'(L)=0. At x=L/2, y, y', y'' and y''' are continuous.

Problem-solving procedure

Step 1. Integrate EI y1^(4)=w0(1-2x/L) four times on 0<=x<=L/2.

Step 2. Integrate EI y2^(4)=0 four times on L/2<=x<=L.

Step 3. Use the four fixed-end conditions.

Step 4. Use four continuity conditions at x=L/2.

Step 5. Solve the eight constants.

The resulting deflection is:

```
For 0 <= x <= L/2:

y1(x) = (w0/(EI)) [
          23 L^2 x^2 /1920
        - 3 L x^3 /80
        + x^4/24
        - x^5/(60L) ]

For L/2 <= x <= L:

y2(x) = (w0/(EI)) [
        - L^4/1920
        + L^3 x/192
```

```
- 17 L^2 x^2/1920
+ L x^3/240 ]
```

Beam deflection y(x) is the piecewise function above. If your sign convention takes downward deflection as negative, multiply the result by -1.

Exam tip: For beam problems, write the support boundary conditions before doing algebra. They are as important as the differential equation.

Question Set 7 - Vector Algebra, Moment and Conservative Fields

7(a) Scalar triple product and linear dependence

Geometrically, |A . (B x C)| is the volume of the parallelepiped formed by A, B and C. If the scalar triple product is zero, the three vectors lie in one plane and are linearly dependent.

```
A = (1,-3,2),  B=(3,2,-1),  C=(2,1,-3)

A . (B x C)
= det [ 1  -3   2
        3   2  -1
        2   1  -3 ]
= -28.
```

Since -28 != 0, the vectors are linearly independent. The parallelepiped volume is 28.

7(b) Moment of a force about a point

Moment about point O is M = r x F, where r goes from O (the moment point) to the point of application of the force.

```
Force F = (3,2,-4)
Application point A = (1,-1,2)
Moment point O = (2,-1,3)

r = A - O = (-1,0,-1)

M = r x F
  = (2,-7,-2).
```

Moment = 2 i - 7 j - 2 k.

7(c) Solenoidal? Conservative? Scalar potential

```
F = (z^2 + 2x + 3y)i
  + (3x + 2y + z)j
  + (y + 2zx)k
```

A field is solenoidal if div F=0 and conservative (in a simply connected region) if curl F=0.

```
div F = d/dx(z^2+2x+3y)
      + d/dy(3x+2y+z)
      + d/dz(y+2zx)
      = 2 + 2 + 2x
      = 4 + 2x != 0.

So F is NOT solenoidal.

curl F = 0,
so F is conservative.
```

Find potential phi from grad(phi)=F. Integrate the i-component with respect to x:

```
phi = x z^2 + x^2 + 3xy + g(y,z).

Compare phi_y with 3x+2y+z:
g_y = 2y+z -> g = y^2 + yz + h(z).

Compare phi_z with y+2zx:
h'(z)=0.
```

A scalar potential is phi = xz^2 + x^2 + 3xy + y^2 + yz + C.

Question Set 8 - Orthogonal Surfaces, Green's Theorem and Flux

8(a) Find a and b for orthogonal surfaces

```
Surface 1:  a x^2 - b y z = a + z
Surface 2:  4x^2 y + z^3 = 4
Point: (1,-1,2)
```

Two surfaces are orthogonal at an intersection point when their normal vectors (gradients) are perpendicular.

Problem-solving procedure

Step 1. First force the point to lie on Surface 1 to obtain one relation between a and b.

Step 2. Compute gradients of both implicit surface functions.

Step 3. Evaluate gradients at the point.

Step 4. Set their dot product equal to zero.

```
Surface 1 as F = a x^2 - b y z - a - z = 0.
At (1,-1,2): a + 2b - a - 2 = 0
=> b=1.

grad F = (2ax, -bz, -by-1)
At the point with b=1:
grad F = (2a, -2, 0).

Surface 2 as G = 4x^2 y + z^3 - 4 = 0.
grad G = (8xy, 4x^2, 3z^2)
At (1,-1,2): grad G = (-8,4,12).

Orthogonality:
(2a,-2,0).(-8,4,12)=0
-16a - 8 = 0
=> a=-1/2.
```

a = -1/2, b = 1.

8(b) Closed line integral around the directed triangle

```
F = P i + Q j
P = 2x + y^2
Q = 3y - 4x
Triangle: (0,0), (2,0), (2,1), counterclockwise as indicated.
```

Green's theorem for positive (counterclockwise) orientation:

```
Integral_C P dx + Q dy
= double integral_R (Q_x - P_y) dA.

Q_x - P_y = -4 - 2y.

Region: 0 <= x <= 2, 0 <= y <= x/2.

Integral = integral_0^2 integral_0^(x/2) (-4-2y) dy dx
         = -14/3.
```

The line integral is -14/3 for the counterclockwise direction shown. Reversing the path changes only the sign.

8(c) Flux through the quarter-cylinder surface

```
A = z i + x j - 3y^2 z k
Cylinder: x^2+y^2=16
First octant, 0 <= z <= 5.
```

The surface is the curved quarter-cylinder. Parameterize it by theta and z:

```
r(theta,z) = (4 cos(theta), 4 sin(theta), z)
0 <= theta <= pi/2,  0 <= z <= 5.

Outward vector area element:
n dS = (4 cos(theta), 4 sin(theta), 0) dtheta dz.
```

On the cylinder x=4cos(theta), so

```
A . n dS
= [z, x, -3y^2 z] . [4cos(theta),4sin(theta),0] dtheta dz
= [4z cos(theta) + 16 cos(theta)sin(theta)] dtheta dz.

Flux = integral_0^5 integral_0^(pi/2)
       [4z cos(theta) + 16cos(theta)sin(theta)] dtheta dz
     = 50 + 40
     = 90.
```

Outward flux through the curved first-octant cylinder surface = 90.

Exam tip: For surface flux, parameterize the surface first. Then compute r_u x r_v; this automatically gives both the normal direction and the dS factor.

Rapid Revision - What to Recognize in the Exam

Question pattern	First move
A = symmetric + skew	Immediately write S=(A+A^T)/2 and K=(A-A^T)/2.
Inverse by row operations	Augment [A|I], row-reduce left side to I.
Rank / consistency	Reduce matrix; compare rank(A), rank([A|b]), and number of unknowns.
Normal form	Use row + column operations to reach [I_r 0; 0 0].
Eigenvalue	Solve det(A-lambda I)=0, then (A-lambda I)v=0.
Laplace of t f(t)	Use -dF/ds.
Integral from 0 to t	Its transform is F(s)/s.
Convolution	Factor F(s)G(s), invert each, integrate f(u)g(t-u).
Conservative field	Check curl F=0, then integrate components to find potential.
Solenoidal field	Check div F=0.
Orthogonal surfaces	Gradients are normal vectors; set their dot product to zero.
Closed planar line integral	Green: integral(Pdx+Qdy)=double integral(Q_x-P_y)dA.
Surface flux	Parameterize surface and use (r_u x r_v) dudv.

Final answers at a glance

Part	Answer
1(b)	S=[[1,3,5/2],[3,2,5/2],[5/2,5/2,-3]], K=[[0,-2,1/2],[2,0,7/2],[-1/2,-7/2,0]]
1(c)	A^-1=[[-1/4,3/4,1/2],[1/8,-3/8,1/4],[1/8,5/8,1/4]]
2(a)	rank=3; normal form=[I3 | 0]
2(b)	P=(1/22)[[3,-7,9],[-1,-5,-3],[5,3,-7]]; Q as shown; PAQ=[I3|0]
3(a)	a=-3 no solution; a=2 infinite; otherwise unique
3(b)	x1=11/23-35t/23, x2=-48/23+44t/23, x3=-40/23+29t/23, x4=t
4(c)	eigenvalues 4,-2,-2; for lambda=4, v=(1,1,2)^T
5(a)	sqrt(pi/s), s>0
5(b)	atan(1/s)/s + ((s-2)^2-4)/(((s-2)^2+4)^2), s>2
5(c)	t-2+(t+2)e^-t
6(a)	x=e^t(t^2/2-3t+2)
6(b)	piecewise polynomial given in Set 6
7(a)	scalar triple product=-28 -> independent
7(b)	M=(2,-7,-2)
7(c)	not solenoidal; conservative; phi=xz^2+x^2+3xy+y^2+yz+C
8(a)	a=-1/2, b=1
8(b)	-14/3 (CCW)
8(c)	90 (outward curved-surface flux)

Study priority for a beginner: First master Question Sets 1-4 (matrix methods) and Set 5 (Laplace identities). Then practice vector-calculus recognition in Sets 7-8. Beam deflection in Set 6 is longer, so learn the boundary-condition structure even if you cannot yet reproduce every algebraic constant.